

2-Stage PCA

1 Two-Stage PCA on Futures-Curve Levels

1.1 General Principle of the Two-Stage PCA

The purpose of the two-stage PCA is to estimate stable common factors while preventing high-volatility maturities from dominating the factor extraction.

The procedure contains two distinct stages.

1. A first PCA with relatively fast exponential time decay is used to estimate the idiosyncratic volatility of each maturity.
2. A second PCA with slower exponential time decay is performed after scaling each maturity by the inverse of its estimated idiosyncratic volatility.

The first stage therefore estimates the relative amount of residual noise associated with each maturity. The second stage extracts the final curve factors from volatility-normalized data.

For futures returns, the procedure may be written schematically as

$$R_t \longrightarrow \text{fast weighted PCA} \longrightarrow \hat{\sigma}_{\varepsilon,t} \longrightarrow \text{volatility rescaling} \longrightarrow \text{slow weighted PCA}.$$

When the input consists of price levels rather than returns, an additional step is necessary. Price levels have a non-zero and time-varying mean, so the data must be centered before each weighted PCA:

$$\log P \longrightarrow \text{weighted centering} \longrightarrow \text{fast weighted PCA} \longrightarrow \hat{\sigma}_{\varepsilon,t} \longrightarrow \text{weighted centering} + \text{volatility rescaling} \longrightarrow \text{slow weighted PCA}.$$

Importantly, the relevant mean is not an arbitrary unconditional mean. At each estimation date, the mean must be computed using only observations in the past rolling window and with the same time weights as the corresponding PCA stage.

1.2 Data and Dimensions

At estimation date t , only information available strictly before t is used. Define the rolling estimation window

$$\mathcal{W}_t = \{t - L, \dots, t - 1\},$$

where:

- L is the number of observations in the rolling window;
- $N = 24$ is the number of constant-maturity contracts, from M1 to M24;
- K is the number of final factors retained, for example $K = 3$.

For each date $u \in \mathcal{W}_t$, define the log-price curve

$$x_u = \begin{pmatrix} \log P_{u,1} \\ \log P_{u,2} \\ \vdots \\ \log P_{u,N} \end{pmatrix} \in \mathbb{R}^N,$$

where $P_{u,i}$ denotes the price at date u of constant maturity i .

The estimation matrix is

$$X_t = \begin{pmatrix} x_{t-L}^\top \\ x_{t-L+1}^\top \\ \vdots \\ x_{t-1}^\top \end{pmatrix} \in \mathbb{R}^{L \times N}.$$

For the TTF M1–M24 curve,

$$X_t \in \mathbb{R}^{L \times 24}.$$

1.3 Stage 1: Fast PCA and Idiosyncratic Volatility Estimation

1.3.1 Fast Exponential Time Weights

Let $\lambda_f \in (0, 1)$ denote the fast exponential-decay parameter. For $u \in \mathcal{W}_t$, define

$$\tilde{w}_{t,u}^f = \lambda_f^{t-1-u}.$$

The normalized weights are

$$w_{t,u}^f = \frac{\lambda_f^{t-1-u}}{\sum_{v=t-L}^{t-1} \lambda_f^{t-1-v}},$$

so that

$$\sum_{u=t-L}^{t-1} w_{t,u}^f = 1.$$

Collect the weights in the vector

$$w_t^f = \begin{pmatrix} w_{t,t-L}^f \\ \vdots \\ w_{t,t-1}^f \end{pmatrix} \in \mathbb{R}^L,$$

and define the diagonal weight matrix

$$W_t^f = \text{diag}(w_{t,t-L}^f, \dots, w_{t,t-1}^f) \in \mathbb{R}^{L \times L}.$$

A smaller value of λ_f , or equivalently a shorter half-life, gives greater importance to recent observations.

The relationship between the decay parameter and the half-life h_f is

$$\lambda_f = 2^{-1/h_f}.$$

1.3.2 Fast Weighted Mean

Because the inputs are log-price levels, the data must first be centered.

The fast weighted mean curve is

$$\mu_t^f = X_t^\top w_t^f \in \mathbb{R}^N.$$

For maturity i ,

$$\mu_{t,i}^f = \sum_{u=t-L}^{t-1} w_{t,u}^f \log P_{u,i}.$$

The centered data matrix is

$$X_t^{f,c} = X_t - \mathbf{1}_L (\mu_t^f)^\top \in \mathbb{R}^{L \times N},$$

where

$$\mathbf{1}_L = \begin{pmatrix} 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^L.$$

The weighted centering condition is

$$\left(X_t^{\text{f},c}\right)^\top w_t^{\text{f}} = 0.$$

Thus, the mean is removed before applying either the time weights or the cross-sectional volatility scaling.

1.3.3 Fast Weighted PCA

Apply the square root of the time weights to the rows:

$$\tilde{X}_t^{\text{f}} = \left(W_t^{\text{f}}\right)^{1/2} X_t^{\text{f},c} \in \mathbb{R}^{L \times N}.$$

Equivalently, the weighted covariance matrix is

$$C_t^{\text{f}} = \left(X_t^{\text{f},c}\right)^\top W_t^{\text{f}} X_t^{\text{f},c} \in \mathbb{R}^{N \times N}.$$

Compute the singular-value decomposition

$$\tilde{X}_t^{\text{f}} = U_t^{\text{f}} S_t^{\text{f}} \left(V_t^{\text{f}}\right)^\top.$$

Assuming $L \geq N$, the dimensions are

$$U_t^{\text{f}} \in \mathbb{R}^{L \times N}, \quad S_t^{\text{f}} \in \mathbb{R}^{N \times N}, \quad V_t^{\text{f}} \in \mathbb{R}^{N \times N}.$$

The columns of V_t^{f} are the eigenvectors of the weighted covariance matrix C_t^{f} . Retain the first K_1 preliminary factors:

$$V_{t,K_1}^{\text{f}} = \begin{pmatrix} v_{t,1}^{\text{f}} & \cdots & v_{t,K_1}^{\text{f}} \end{pmatrix} \in \mathbb{R}^{N \times K_1}.$$

For example, one may use

$$K_1 = 3.$$

1.3.4 Stage-1 Residuals

The factor scores for the observations in the estimation window are

$$F_t^{\text{f}} = X_t^{\text{f},c} V_{t,K_1}^{\text{f}} \in \mathbb{R}^{L \times K_1}.$$

The fitted centered log-price curves are

$$\hat{X}_t^{\text{f},c} = F_t^{\text{f}} \left(V_{t,K_1}^{\text{f}}\right)^\top.$$

Equivalently,

$$\hat{X}_t^{\text{f},c} = X_t^{\text{f},c} V_{t,K_1}^{\text{f}} \left(V_{t,K_1}^{\text{f}}\right)^\top \in \mathbb{R}^{L \times N}.$$

The stage-1 residual matrix is

$$E_t^{\text{f}} = X_t^{\text{f},c} - \hat{X}_t^{\text{f},c} \in \mathbb{R}^{L \times N}.$$

For maturity i , estimate the fast weighted idiosyncratic variance as

$$\hat{\sigma}_{\varepsilon,t,i}^2 = \sum_{u=t-L}^{t-1} w_{t,u}^f \left(E_{t,u,i}^f \right)^2.$$

Because the residuals are weighted-centered, no additional residual mean needs to be subtracted. The corresponding idiosyncratic volatility is

$$\hat{\sigma}_{\varepsilon,t,i} = \sqrt{\hat{\sigma}_{\varepsilon,t,i}^2}.$$

Collect these estimates in

$$\hat{\sigma}_{\varepsilon,t} = \begin{pmatrix} \hat{\sigma}_{\varepsilon,t,1} \\ \vdots \\ \hat{\sigma}_{\varepsilon,t,N} \end{pmatrix} \in \mathbb{R}^N,$$

and define

$$D_{\varepsilon,t} = \text{diag}(\hat{\sigma}_{\varepsilon,t,1}, \dots, \hat{\sigma}_{\varepsilon,t,N}) \in \mathbb{R}^{N \times N}.$$

To prevent an excessively large inverse-volatility weight, impose a volatility floor:

$$\hat{\sigma}_{\varepsilon,t,i} \leftarrow \max(\hat{\sigma}_{\varepsilon,t,i}, \sigma_{\min}).$$

One may alternatively use a shrinkage estimator,

$$\tilde{\sigma}_{\varepsilon,t,i}^2 = (1 - \alpha)\hat{\sigma}_{\varepsilon,t,i}^2 + \alpha\bar{\sigma}_{\varepsilon,t}^2,$$

where

$$\bar{\sigma}_{\varepsilon,t}^2 = \frac{1}{N} \sum_{i=1}^N \hat{\sigma}_{\varepsilon,t,i}^2.$$

1.4 Stage 2: Slow Volatility-Rewighted PCA

1.4.1 Slow Exponential Time Weights

Let $\lambda_s \in (0, 1)$ denote the slow decay parameter, with

$$\lambda_s > \lambda_f.$$

Equivalently, the slow half-life satisfies

$$h_s > h_f.$$

For $u \in \mathcal{W}_t$, define

$$w_{t,u}^s = \frac{\lambda_s^{t-1-u}}{\sum_{v=t-L}^{t-1} \lambda_s^{t-1-v}}.$$

Therefore,

$$\sum_{u=t-L}^{t-1} w_{t,u}^s = 1.$$

Define

$$w_t^s \in \mathbb{R}^L$$

and

$$W_t^s = \text{diag}(w_{t,t-L}^s, \dots, w_{t,t-1}^s) \in \mathbb{R}^{L \times L}.$$

1.4.2 Slow Weighted Mean

The final PCA uses its own weighted mean, computed consistently with the slow time weights:

$$\mu_t^s = X_t^\top w_t^s \in \mathbb{R}^N.$$

For each maturity i ,

$$\mu_{t,i}^s = \sum_{u=t-L}^{t-1} w_{t,u}^s \log P_{u,i}.$$

The slow-centered data matrix is

$$X_t^{s,c} = X_t - \mathbf{1}_L (\mu_t^s)^\top \in \mathbb{R}^{L \times N}.$$

It satisfies

$$(X_t^{s,c})^\top w_t^s = 0.$$

The mean μ_t^s is the mean curve that must be added back when reconstructing the final conditional curve.

1.4.3 Idiosyncratic-Volatility Rescaling

Each maturity is divided by its stage-1 idiosyncratic volatility:

$$Z_t = X_t^{s,c} D_{\varepsilon,t}^{-1} \in \mathbb{R}^{L \times N}.$$

Element by element,

$$Z_{t,u,i} = \frac{\log P_{u,i} - \mu_{t,i}^s}{\hat{\sigma}_{\varepsilon,t,i}}.$$

The purpose of this transformation is to prevent a maturity with high idiosyncratic variation from dominating the final PCA.

The slow time weights are then applied:

$$\tilde{Z}_t = (W_t^s)^{1/2} Z_t.$$

Therefore,

$$\tilde{Z}_t = (W_t^s)^{1/2} \left[X_t - \mathbf{1}_L (\mu_t^s)^\top \right] D_{\varepsilon,t}^{-1}$$

with

$$\tilde{Z}_t \in \mathbb{R}^{L \times N}.$$

The order of operations is therefore

$$\boxed{\text{log-prices} \rightarrow \text{weighted centering} \rightarrow \text{idiosyncratic-volatility scaling} \rightarrow \text{time weighting} \rightarrow \text{SVD}.}$$

The data should not be centered using an unweighted mean after the volatility rescaling. The correct mean is the slow weighted mean of the original log-price levels, computed before the cross-sectional scaling.

1.4.4 Final SVD

Compute

$$\tilde{Z}_t = U_t S_t V_t^\top.$$

Assuming $L \geq N$,

$$U_t \in \mathbb{R}^{L \times N}, \quad S_t \in \mathbb{R}^{N \times N}, \quad V_t \in \mathbb{R}^{N \times N}.$$

Retain the first K right singular vectors:

$$V_{t,K} = \begin{pmatrix} v_{t,1} & \cdots & v_{t,K} \end{pmatrix} \in \mathbb{R}^{N \times K}.$$

For example, with

$$N = 24, \quad K = 3,$$

we obtain

$$V_{t,K} \in \mathbb{R}^{24 \times 3}.$$

The columns of $V_{t,K}$ are orthonormal in the standardized space:

$$V_{t,K}^\top V_{t,K} = I_K.$$

1.5 Factor Scores and Loadings

The vectors in $V_{t,K}$ are factor loadings in the idiosyncratic-volatility-standardized space.

For any log-price curve x_u , define the centered and standardized curve relative to the model estimated at date t :

$$z_{u|t} = D_{\varepsilon,t}^{-1} (x_u - \mu_t^s) \in \mathbb{R}^N.$$

The corresponding factor-score vector is

$$f_{u|t} = V_{t,K}^\top z_{u|t}.$$

Therefore,

$$f_{u|t} = V_{t,K}^\top D_{\varepsilon,t}^{-1} (x_u - \mu_t^s)$$

with

$$f_{u|t} \in \mathbb{R}^K.$$

If $K = 3$,

$$f_{u|t} = \begin{pmatrix} f_{u|t,1} \\ f_{u|t,2} \\ f_{u|t,3} \end{pmatrix}.$$

The factors will typically correspond approximately to:

- a level factor;
- a slope factor;
- a curvature or seasonal-shape factor.

However, because the PCA is estimated on volatility-standardized log-prices, the factor interpretation must be verified from the empirical loading shapes rather than imposed ex ante.

1.6 Curve Reconstruction

In the standardized space, the K -factor reconstruction is

$$\hat{z}_{u|t} = V_{t,K} f_{u|t} \in \mathbb{R}^N.$$

Returning to the original log-price space gives

$$\hat{x}_{u|t} = \mu_t^s + D_{\varepsilon,t} \hat{z}_{u|t}.$$

Hence,

$$\boxed{\hat{x}_{u|t} = \mu_t^s + D_{\varepsilon,t} V_{t,K} f_{u|t}}$$

Define the factor loading matrix in the original log-price space as

$$B_t = D_{\varepsilon,t} V_{t,K} \in \mathbb{R}^{N \times K}.$$

The factor representation is then

$$\hat{x}_{u|t} = \mu_t^s + B_t f_{u|t}.$$

The residual log-price curve is

$$\varepsilon_{u|t} = x_u - \hat{x}_{u|t}.$$

Equivalently,

$$x_u = \mu_t^s + B_t f_{u|t} + \varepsilon_{u|t}.$$

Finally, the reconstructed futures prices are

$$\hat{P}_{u,i|t} = \exp(\hat{x}_{u,i|t}), \quad i = 1, \dots, N.$$

1.7 Conditional Curve Based on Fundamentals

Let

$$q_u \in \mathbb{R}^J$$

denote the vector of fundamental variables observed at date u . These may include storage, LNG sendout, pipeline flows, weather forecasts, power generation, outages, and cross-commodity variables.

The relationship between the PCA scores and the fundamentals may be estimated as

$$f_{u|t} = g(q_u) + \eta_u,$$

where

$$g : \mathbb{R}^J \rightarrow \mathbb{R}^K.$$

Alternatively, each factor may be modelled separately:

$$f_{u|t,k} = g_k(q_u) + \eta_{u,k}, \quad k = 1, \dots, K.$$

At date t , the predicted factor vector is

$$\hat{f}_t^{\text{fund}} = \hat{g}(q_t) \in \mathbb{R}^K.$$

The conditional log-price curve implied by the fundamentals is

$$\boxed{\hat{x}_t^{\text{fund}} = \mu_t^s + D_{\varepsilon,t} V_{t,K} \hat{f}_t^{\text{fund}}}$$

or equivalently,

$$\hat{x}_t^{\text{fund}} = \mu_t^s + B_t \hat{f}_t^{\text{fund}}.$$

The corresponding conditional futures-price curve is

$$\hat{P}_{t,i}^{\text{fund}} = \exp\left(\hat{x}_{t,i}^{\text{fund}}\right).$$

The maturity-specific deviation of the observed market curve from the fundamental curve is

$$d_{t,i} = \log P_{t,i}^{\text{market}} - \hat{x}_{t,i}^{\text{fund}}.$$

A positive value,

$$d_{t,i} > 0,$$

indicates that maturity i is trading above the level implied by the fundamental model, whereas

$$d_{t,i} < 0$$

indicates that it is trading below the model-implied level.

For small deviations, $d_{t,i}$ is approximately the percentage mispricing:

$$\frac{P_{t,i}^{\text{market}} - \hat{P}_{t,i}^{\text{fund}}}{\hat{P}_{t,i}^{\text{fund}}} \approx d_{t,i}.$$

1.8 Choice of the Mean for Price-Level PCA

The centering mean must satisfy three conditions.

First, it must be maturity-specific:

$$\mu_t = \begin{pmatrix} \mu_{t,1} \\ \vdots \\ \mu_{t,N} \end{pmatrix},$$

rather than a single scalar mean applied to the full curve.

Second, it must be estimated using only observations available before date t :

$$\mu_t = \mu_t(x_{t-L}, \dots, x_{t-1}).$$

Third, it must use the same temporal weights as the PCA stage in which it appears.

Thus, the fast stage uses

$$\mu_t^f = X_t^\top w_t^f,$$

whereas the slow stage uses

$$\mu_t^s = X_t^\top w_t^s.$$

Using a single unweighted mean for both stages would be inconsistent with the exponentially weighted covariance matrices. Similarly, using a full-sample mean would introduce look-ahead bias.

The final reconstruction must use the slow mean

$$\mu_t^s,$$

because the final factors $V_{t,K}$ are estimated from the slow-centered data.

1.9 Comparison with the Return-Based Version

For log-returns,

$$r_u = x_u - x_{u-1},$$

the unconditional mean is generally small relative to return volatility. It is therefore common in practice to approximate

$$\mathbb{E}[r_u] \approx 0$$

and omit explicit centering.

Strictly speaking, a weighted PCA on returns should nevertheless use the weighted return mean

$$\bar{r}_t = R_t^\top w_t.$$

The centered returns are

$$R_t^c = R_t - \mathbf{1}_L \bar{r}_t^\top.$$

On daily futures returns, the numerical difference may be small. On log-price levels, however, omitting the mean is not a reasonable approximation because

$$\mathbb{E}[\log P_{t,i}] \neq 0.$$

Without centering, the first singular vector would largely reflect the absolute location of the data relative to the origin, rather than variation of the futures curve around its locally estimated mean.

1.10 Compact Two-Stage Formulation

The complete stage-1 transformation is

$$\boxed{\tilde{X}_t^f = \left(W_t^f\right)^{1/2} \left[X_t - \mathbf{1}_L \left(\mu_t^f\right)^\top\right]}$$

with

$$\mu_t^f = X_t^\top w_t^f.$$

The stage-1 residuals provide

$$D_{\varepsilon,t} = \text{diag}(\hat{\sigma}_{\varepsilon,t,1}, \dots, \hat{\sigma}_{\varepsilon,t,N}).$$

The complete stage-2 transformation is

$$\boxed{\tilde{Z}_t = (W_t^s)^{1/2} \left[X_t - \mathbf{1}_L (\mu_t^s)^\top\right] D_{\varepsilon,t}^{-1}}$$

with

$$\mu_t^s = X_t^\top w_t^s.$$

After the final SVD,

$$\tilde{Z}_t = U_t S_t V_t^\top,$$

the conditional curve based on predicted fundamental factors is

$$\boxed{\widehat{\log P_t}^{\text{fund}} = \mu_t^s + D_{\varepsilon,t} V_{t,K} \hat{f}_t^{\text{fund}}.}$$

1.11 Seasonality Adjustment

Natural gas forward curves exhibit a strong deterministic seasonal component driven by the annual heating and cooling cycles. Since the objective is to model the dynamic component of the curve rather than its average seasonal shape, we also consider a deseasonalized version of the forward curve.

Rather than removing the average price level of each delivery month, which would also remove long-term changes in the gas price level, we estimate a *relative seasonal curve*. The idea is to isolate the typical winter–summer shape while preserving the overall level of the curve.

For each historical date u , define the log-price curve

$$x_u = \begin{pmatrix} \log P_{u,1} \\ \vdots \\ \log P_{u,N} \end{pmatrix},$$

and compute its cross-sectional level

$$\ell_u = \frac{1}{N} \sum_{i=1}^N x_{u,i}.$$

The relative curve is then

$$r_{u,i} = x_{u,i} - \ell_u.$$

Hence,

$$r_{u,i} > 0$$

indicates that maturity i is expensive relative to the average curve level on date u , while

$$r_{u,i} < 0$$

indicates that it is relatively cheap.

Let

$$c(u, i) \in \{1, \dots, 12\}$$

denote the calendar delivery month associated with maturity i on date u .

Using only the historical observations available before the estimation date t , the seasonal effect of each delivery month is estimated as

$$\tilde{\gamma}_{t,m} = \text{median} \{r_{u,i} : c(u, i) = m, u \in [t - H, t - 1]\},$$

where H denotes the historical estimation window (for example five years).

The median is preferred to the mean because it is substantially more robust to extreme gas-price episodes such as the 2021–2022 energy crisis.

The seasonal coefficients are then centered to ensure that they only represent a relative curve shape,

$$\gamma_{t,m} = \tilde{\gamma}_{t,m} - \frac{1}{12} \sum_{j=1}^{12} \tilde{\gamma}_{t,j}.$$

At date t , the seasonal vector associated with the observed forward curve is therefore

$$s_t = \begin{pmatrix} \gamma_{t,c(t,1)} \\ \vdots \\ \gamma_{t,c(t,N)} \end{pmatrix}.$$

For numerical stability, the vector is finally recentered,

$$s_t \leftarrow s_t - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top s_t,$$

so that

$$\frac{1}{N} \sum_{i=1}^N s_{t,i} = 0.$$

The deseasonalized log-price curve is then defined as

$$y_t = x_t - s_t.$$

The complete PCA procedure is applied to the deseasonalized curves $\{y_t\}$ instead of the raw log-prices.

After the factors have been estimated and the curve reconstructed,

$$\hat{y}_t = \mu_t + D_{\varepsilon,t} V_{t,K} \hat{f}_t,$$

the seasonal component is added back,

$$\hat{x}_t = s_t + \hat{y}_t,$$

and the reconstructed futures prices are finally

$$\hat{P}_{t,i} = \exp(\hat{x}_{t,i}).$$

Notice that this procedure does *not* remove the overall level of the forward curve from the PCA. The cross-sectional level ℓ_u is only used to estimate the seasonal template $\{\gamma_m\}$. The PCA itself is performed on $y_t = x_t - s_t$, whose average level remains unchanged because the seasonal vector has zero cross-sectional mean.

1.12 Seasonality Adjustment and Order of Operations

The seasonal adjustment must be performed before the rolling centering used in the two-stage PCA.

The complete preprocessing order is

$P_{u,i} \rightarrow x_{u,i} = \log P_{u,i} \rightarrow \hat{s}_{u,i} \rightarrow y_{u,i} = x_{u,i} - \hat{s}_{u,i} \rightarrow \text{rolling weighted centering} \rightarrow \text{idiosyncratic-volatility scaling}$
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The seasonal vector is estimated separately from the PCA transformation. In particular, the cross-sectional level is removed only to estimate the typical relative seasonal shape.

1.12.1 Step 1: Raw Log-Price Curves

For each historical date u , define the log-price curve

$$x_u = \begin{pmatrix} \log P_{u,1} \\ \vdots \\ \log P_{u,N} \end{pmatrix} \in \mathbb{R}^N.$$

For the TTF M1–M24 curve,

$$N = 24.$$

1.12.2 Step 2: Remove the Cross-Sectional Level Only for Seasonal Estimation

For each date u , compute the cross-sectional curve level

$$\ell_u = \frac{1}{N} \sum_{i=1}^N x_{u,i}.$$

Define the relative curve

$$r_{u,i} = x_{u,i} - \ell_u.$$

Equivalently,

$$r_u = x_u - \ell_u \mathbf{1}_N.$$

This operation is used only to estimate the seasonal shape. The matrix $\{r_u\}$ is not the final input to the PCA.

The purpose is to prevent the seasonal estimator from confusing the winter–summer shape with the general level of European gas prices.

1.12.3 Step 3: Estimate the Seasonal Effect by Delivery Month

Let

$$c(u, i) \in \{1, \dots, 12\}$$

denote the calendar delivery month associated with maturity i at date u .

At estimation date t , using only observations available before t , estimate the raw seasonal effect for delivery month m as

$$\tilde{\gamma}_{t,m} = \text{median} \{r_{u,i} : c(u, i) = m, u \in [t - H, t - 1]\},$$

where H is the seasonal estimation window, for example five years.

A mean-based alternative is

$$\tilde{\gamma}_{t,m}^{\text{mean}} = \frac{\sum_{u,i} r_{u,i} \mathbf{1}_{\{c(u,i)=m\}}}{\sum_{u,i} \mathbf{1}_{\{c(u,i)=m\}}}.$$

The median estimator is more robust to extreme price episodes, whereas the mean estimator is smoother and fully linear. Both can be compared as a robustness check.

The twelve seasonal coefficients are centered as

$$\gamma_{t,m} = \tilde{\gamma}_{t,m} - \frac{1}{12} \sum_{j=1}^{12} \tilde{\gamma}_{t,j},$$

so that

$$\frac{1}{12} \sum_{m=1}^{12} \gamma_{t,m} = 0.$$

1.12.4 Step 4: Construct the Seasonal Vector for Each Curve

For any date u , the seasonal vector implied by the delivery months of the observed contracts is

$$s_{u|t} = \begin{pmatrix} \gamma_{t,c(u,1)} \\ \vdots \\ \gamma_{t,c(u,N)} \end{pmatrix} \in \mathbb{R}^N.$$

The notation $s_{u|t}$ indicates that the seasonal coefficients are estimated using information available at estimation date t , and then mapped to the delivery months present in curve u .

To guarantee that the seasonal adjustment does not change the cross-sectional level of the curve, recenter the seasonal vector as

$$s_{u|t} \leftarrow s_{u|t} - \frac{1}{N} \mathbf{1}_N \mathbf{1}_N^\top s_{u|t}.$$

Hence,

$$\frac{1}{N} \sum_{i=1}^N s_{u|t,i} = 0.$$

1.12.5 Step 5: Remove Seasonality Before PCA Centering

The deseasonalized log-price curve is

$$y_{u|t} = x_u - s_{u|t}.$$

Therefore, the sequence is

$$\boxed{x_u \rightarrow s_{u|t} \rightarrow y_{u|t} = x_u - s_{u|t}.$$

Only after this seasonal adjustment is the rolling PCA centering applied. For the rolling window

$$\mathcal{W}_t = \{t - L, \dots, t - 1\},$$

define the deseasonalized data matrix

$$Y_t = \begin{pmatrix} y_{t-L|t}^\top \\ \vdots \\ y_{t-1|t}^\top \end{pmatrix} \in \mathbb{R}^{L \times N}.$$

The two-stage PCA is then applied to Y_t , not to the raw log-price matrix X_t .

1.12.6 Step 6: Stage-1 Fast Weighted Centering

The fast weighted mean of the deseasonalized curves is

$$\mu_{Y,t}^f = Y_t^\top w_t^f \in \mathbb{R}^N.$$

The fast-centered matrix is

$$Y_t^{f,c} = Y_t - \mathbf{1}_L \left(\mu_{Y,t}^f \right)^\top.$$

The stage-1 weighted PCA matrix is

$$\tilde{Y}_t^f = \left(W_t^f \right)^{1/2} Y_t^{f,c}.$$

Thus, the stage-1 order is

raw log-prices $\rightarrow$ remove seasonality $\rightarrow$ fast weighted centering $\rightarrow$ fast time weighting $\rightarrow$ stage-1 PCA.

The stage-1 PCA residuals are then used to estimate the idiosyncratic volatility matrix

$$D_{\varepsilon,t} = \text{diag}(\hat{\sigma}_{\varepsilon,t,1}, \dots, \hat{\sigma}_{\varepsilon,t,N}).$$

1.12.7 Step 7: Stage-2 Slow Centering and Volatility Rescaling

The slow weighted mean is also computed from the deseasonalized data:

$$\mu_{Y,t}^s = Y_t^\top w_t^s \in \mathbb{R}^N.$$

The slow-centered matrix is

$$Y_t^{s,c} = Y_t - \mathbf{1}_L \left(\mu_{Y,t}^s \right)^\top.$$

The idiosyncratic-volatility-scaled matrix is

$$Z_t = Y_t^{s,c} D_{\varepsilon,t}^{-1}.$$

The slow exponential time weights are then applied:

$$\tilde{Z}_t = (W_t^s)^{1/2} Z_t.$$

Equivalently,

$$\tilde{Z}_t = (W_t^s)^{1/2} \left[Y_t - \mathbf{1}_L \left(\mu_{Y,t}^s \right)^\top \right] D_{\varepsilon,t}^{-1}.$$

Since

$$Y_t = X_t - S_t,$$

the complete stage-2 transformation is

$$\tilde{Z}_t = (W_t^s)^{1/2} \left[X_t - S_t - \mathbf{1}_L \left(\mu_{Y,t}^s \right)^\top \right] D_{\varepsilon,t}^{-1}.$$

The exact stage-2 order is therefore

log-prices $\rightarrow$ remove seasonality $\rightarrow$ slow weighted centering $\rightarrow$ divide by idiosyncratic volatility $\rightarrow$ apply slow ti

1.12.8 Step 8: Factor Estimation

Compute the final singular-value decomposition

$$\tilde{Z}_t = U_t S_t V_t^\top.$$

Retain the first K right singular vectors

$$V_{t,K} \in \mathbb{R}^{N \times K}.$$

For a deseasonalized curve y_u , the factor-score vector is

$$f_{u|t} = V_{t,K}^\top D_{\varepsilon,t}^{-1} \left(y_u - \mu_{Y,t}^s \right).$$

Since

$$y_u = x_u - s_{u|t},$$

this can also be written as

$$f_{u|t} = V_{t,K}^\top D_{\varepsilon,t}^{-1} (x_u - s_{u|t} - \mu_{Y,t}^s).$$

1.12.9 Step 9: Reconstruction in Reverse Order

The predicted deseasonalized log-price curve is

$$\hat{y}_t = \mu_{Y,t}^s + D_{\varepsilon,t} V_{t,K} \hat{f}_t.$$

The seasonal component is then added back:

$$\boxed{\hat{x}_t = s_t + \hat{y}_t.}$$

Hence,

$$\boxed{\hat{x}_t = s_t + \mu_{Y,t}^s + D_{\varepsilon,t} V_{t,K} \hat{f}_t.}$$

Finally,

$$\hat{P}_{t,i} = \exp(\hat{x}_{t,i}).$$

The reconstruction order is therefore the reverse of the preprocessing order:

$$\boxed{\hat{f}_t \rightarrow \text{PCA reconstruction in standardized space} \rightarrow \text{restore idiosyncratic-volatility scale} \rightarrow \text{add the rolling mean}}$$

More explicitly,

$$\boxed{\hat{f}_t \rightarrow V_{t,K} \hat{f}_t \rightarrow D_{\varepsilon,t} V_{t,K} \hat{f}_t \rightarrow \mu_{Y,t}^s + D_{\varepsilon,t} V_{t,K} \hat{f}_t \rightarrow s_t + \mu_{Y,t}^s + D_{\varepsilon,t} V_{t,K} \hat{f}_t.}$$

1.12.10 Important Distinction Between the Two Centering Operations

Two different level-removal operations appear in the procedure.

First,

$$r_{u,i} = x_{u,i} - \ell_u$$

is used only to estimate the seasonal template. This cross-sectional demeaning is temporary and is not the input to the PCA.

Second,

$$Y_t^c = Y_t - \mathbf{1}_L \mu_{Y,t}^\top$$

is the actual rolling time-series centering used by the PCA.

Therefore,

$$\boxed{\text{cross-sectional demeaning} \neq \text{rolling PCA centering.}}$$

The first isolates the typical winter–summer shape. The second removes the rolling time-series mean of each maturity before the covariance matrix is estimated.

1.13 Model Comparison

Three alternative specifications will be compared.

- **Model A:** Two-stage PCA on raw log-prices with $K = 3$ factors.
- **Model B:** Two-stage PCA on raw log-prices with $K = 4$ factors.
- **Model C:** Two-stage PCA on deseasonalized log-prices with $K = 3$ factors.

For Model C, the decomposition is

$$y_t = x_t - s_t,$$

followed by the reconstruction

$$\hat{x}_t = s_t + \hat{y}_t.$$

The competing models are evaluated out-of-sample using several complementary criteria.

- Root Mean Squared Error (RMSE) on reconstructed log-prices;
- maturity-by-maturity RMSE;
- reconstruction error of calendar spreads;
- stability of the estimated PCA loadings over time;
- performance of the relative-value signal based on the pricing residual

$$x_t - \hat{x}_t.$$

2 Weekly Valuation Workflow, PCA Diagnostics, and Model Selection

2.1 Objective

The objective is to build a weekly attribution and valuation model for the TTF forward curve.

At each Friday t , the model uses the fundamental information available before the settlement cutoff to estimate the curve implied by fundamentals. This model-implied curve is then compared with the observed Friday settlement curve.

Let

$$x_t = \begin{pmatrix} \log P_{t,M1} \\ \vdots \\ \log P_{t,M24} \end{pmatrix} \in \mathbb{R}^N, \quad N = 24,$$

denote the observed Friday log-price curve, and let

$$q_t \in \mathbb{R}^J$$

denote the vector of fundamentals available before the Friday cutoff.

The model first represents the curve through a low-dimensional PCA factor structure,

$$x_t = s_t + \mu_t + D_{\varepsilon,t} V_{t,K} f_t + \varepsilon_t,$$

where:

- $s_t \in \mathbb{R}^N$ is the optional seasonal curve;
- $\mu_t \in \mathbb{R}^N$ is the rolling weighted mean curve;
- $D_{\varepsilon,t} \in \mathbb{R}^{N \times N}$ contains the maturity-specific idiosyncratic volatility estimates;
- $V_{t,K} \in \mathbb{R}^{N \times K}$ contains the first K PCA loadings;
- $f_t \in \mathbb{R}^K$ is the observed factor-score vector;

- $\varepsilon_t \in \mathbb{R}^N$ is the PCA reconstruction residual.

The factor scores are then related to the fundamental variables through

$$f_t = g(q_t) + \eta_t.$$

The resulting model-implied curve is

$$\widehat{x}_t^{\text{fund}} = s_t + \mu_t + D_{\varepsilon,t} V_{t,K} \widehat{f}_t,$$

where

$$\widehat{f}_t = \widehat{g}(q_t).$$

The valuation residual is

$$d_t = x_t - \widehat{x}_t^{\text{fund}}.$$

A positive component $d_{t,i} > 0$ indicates that maturity i trades above the level implied by the model, whereas $d_{t,i} < 0$ indicates that it trades below the model-implied level.

2.2 Weekly Walk-Forward Timeline

The model must be evaluated through a strictly chronological walk-forward procedure.

Suppose that t denotes the current Friday and that $t - 1$ denotes the previous Friday.

2.2.1 Step 1: Fit the Model Using Information Available up to Friday $t - 1$

Using only observations available up to and including Friday $t - 1$, estimate

$$s_{\cdot|t-1}, \quad \mu_{t-1}, \quad D_{\varepsilon,t-1}, \quad V_{t-1,K}.$$

The PCA estimation window is

$$\mathcal{W}_{t-1}^{\text{PCA}} = \{t - 1 - L + 1, \dots, t - 1\},$$

where L is the number of weekly observations in the rolling PCA window.

If the deseasonalized specification is used, the input curves are

$$y_u = x_u - s_{u|t-1}.$$

Otherwise,

$$y_u = x_u.$$

The two-stage PCA is estimated from the matrix

$$Y_{t-1} = \begin{pmatrix} y_{t-L}^\top \\ \vdots \\ y_{t-1}^\top \end{pmatrix} \in \mathbb{R}^{L \times N}.$$

2.2.2 Step 2: Construct Historical Factor Targets

Once the current PCA basis has been estimated, every historical curve in the regression training window is projected into the same factor space.

For each historical Friday $u \leq t - 1$,

$$f_{u|t-1} = V_{t-1,K}^\top D_{\varepsilon,t-1}^{-1} (x_u - s_{u|t-1} - \mu_{t-1}).$$

The historical factor-target matrix is

$$F_{\text{train}|t-1} = \begin{pmatrix} f_{u_1|t-1}^\top \\ \vdots \\ f_{u_M|t-1}^\top \end{pmatrix} \in \mathbb{R}^{M \times K}.$$

All historical targets therefore use the same PCA coordinate system estimated at the current refit date.

2.2.3 Step 3: Fit the Fundamental Regression

Using the historical fundamental variables and factor targets, estimate

$$f_{u|t-1} = g_{t-1}(q_u) + \eta_u.$$

Equivalently, each factor may be modelled separately:

$$f_{u|t-1,k} = g_{t-1,k}(q_u) + \eta_{u,k}, \quad k = 1, \dots, K.$$

The complete model estimated at Friday $t - 1$ is

$$\mathcal{M}_{t-1} = (s_{\cdot|t-1}, \mu_{t-1}, D_{\varepsilon,t-1}, V_{t-1,K}, g_{t-1}).$$

This model is frozen until the next weekly refit.

2.2.4 Step 4: Construct the Friday- t Fundamental Vector

During week t , construct the vector

$$q_t = \text{fundamental information available before the Friday cutoff.}$$

The information set may include:

- the latest storage level and storage anomaly;
- the latest storage surprise;
- weekly average and Friday level of LNG sendout;
- Norwegian pipeline flows and outages;
- weather forecasts available before the cutoff;
- wind, power demand, power burn, coal, EUA, JKM, and related cross-commodity variables;
- weekly changes and deviations from seasonal norms.

All features must be timestamped according to their actual publication or availability time.

2.2.5 Step 5: Produce the Friday- t Implied Curve

Using the frozen regression model,

$$\hat{f}_{t|t-1} = g_{t-1}(q_t).$$

The corresponding model-implied log-price curve is

$$\hat{x}_{t|t-1}^{\text{fund}} = s_{t|t-1} + \mu_{t-1} + D_{\varepsilon,t-1} V_{t-1,K} \hat{f}_{t|t-1}.$$

The associated price curve is

$$\hat{P}_{t,i}^{\text{fund}} = \exp\left(\hat{x}_{t,i|t-1}^{\text{fund}}\right).$$

2.2.6 Step 6: Compare with the Observed Friday Settlement

Once the Friday settlement curve x_t is available, compute the valuation residual

$$d_t = x_t - \hat{x}_{t|t-1}^{\text{fund}}.$$

The observed factor scores may also be computed as

$$f_{t|t-1}^{\text{obs}} = V_{t-1,K}^\top D_{\varepsilon,t-1}^{-1} \left(x_t - s_{t|t-1} - \mu_{t-1} \right).$$

The factor-level attribution error is

$$a_t = f_{t|t-1}^{\text{obs}} - \hat{f}_{t|t-1}.$$

Thus, the model provides both:

$$\text{factor attribution} \quad \text{and} \quad \text{maturity-level valuation.}$$

2.2.7 Step 7: Refit After the Friday- t Evaluation

Only after the Friday- t valuation has been produced and evaluated is the new observation added to the estimation sample.

The next model is then estimated using

$$\{x_u, q_u : u \leq t\}.$$

The updated model

$$\mathcal{M}_t$$

is subsequently used to value the curve at Friday $t + 1$.

The complete walk-forward workflow is therefore

$$\begin{aligned} &\text{Fit at } t-1 \rightarrow \text{freeze PCA and regression} \\ &\rightarrow q_t \rightarrow \hat{f}_{t|t-1} \rightarrow \hat{x}_{t|t-1}^{\text{fund}} \\ &\rightarrow x_t - \hat{x}_{t|t-1}^{\text{fund}} \\ &\rightarrow \text{add observation } t \rightarrow \text{refit for } t+1. \end{aligned}$$

2.3 Metrics for Evaluating the PCA

The PCA must first be evaluated independently from the fundamental regression. The objective is to verify that a small number of factors accurately and stably represents the weekly TTF curve.

2.3.1 Explained Variance Ratio

For K retained factors, define

$$\text{EVR}_{t,K} = \frac{\sum_{k=1}^K \lambda_{t,k}}{\sum_{j=1}^N \lambda_{t,j}},$$

where $\lambda_{t,k}$ denotes the k -th PCA eigenvalue.

The following quantities should be monitored:

- average explained variance;
- median explained variance;
- lower-tail explained variance;
- explained variance during stress periods.

A high explained variance is desirable but is not sufficient, especially for a PCA on price levels, where the first factor may mechanically capture a large common price level.

2.3.2 Pure PCA Reconstruction Error

Using the observed factor scores rather than the factor scores predicted by the regression, define

$$\hat{x}_t^{\text{PCA}} = s_t + \mu_t + D_{\varepsilon,t} V_{t,K} f_t^{\text{obs}}.$$

The PCA reconstruction residual is

$$e_t^{\text{PCA}} = x_t - \hat{x}_t^{\text{PCA}}.$$

The global log-price RMSE is

$$\text{RMSE}_{\text{PCA}} = \sqrt{\frac{1}{TN} \sum_{t=1}^T \sum_{i=1}^N (e_{t,i}^{\text{PCA}})^2}.$$

2.3.3 RMSE by Maturity

For each maturity i ,

$$\text{RMSE}_i^{\text{PCA}} = \sqrt{\frac{1}{T} \sum_{t=1}^T (e_{t,i}^{\text{PCA}})^2}.$$

This metric identifies whether the PCA performs poorly on particular parts of the curve, such as the front, the belly, or the back end.

2.3.4 Calendar-Spread Reconstruction Error

For a spread between maturities i and j , define

$$\Delta x_{t,i,j} = x_{t,i} - x_{t,j}.$$

The reconstructed spread is

$$\Delta \hat{x}_{t,i,j}^{\text{PCA}} = \hat{x}_{t,i}^{\text{PCA}} - \hat{x}_{t,j}^{\text{PCA}}.$$

The spread error is

$$e_{t,i,j}^{\text{spread}} = \Delta x_{t,i,j} - \Delta \hat{x}_{t,i,j}^{\text{PCA}}.$$

The spread RMSE is

$$\text{RMSE}_{i,j}^{\text{spread}} = \sqrt{\frac{1}{T} \sum_{t=1}^T \left(e_{t,i,j}^{\text{spread}} \right)^2}.$$

This metric is particularly important because the ultimate objective is a relative-value analysis rather than an outright directional forecast.

Relevant spreads may include

$$M1 - M2, \quad M1 - M3, \quad M3 - M6, \quad M6 - M12, \quad M12 - M24,$$

as well as winter–summer spreads.

2.3.5 Loading Stability

After aligning factor signs, define the loading similarity

$$\rho_{t,k} = \left| v_{t,k}^\top v_{t-1,k} \right|.$$

Values close to one indicate stable factor loadings.

Because individual factors may rotate when eigenvalues are close, the stability of the complete factor subspace should also be measured through

$$d_t^{\text{subspace}} = \left\| V_{t,K} V_{t,K}^\top - V_{t-1,K} V_{t-1,K}^\top \right\|_F.$$

A smaller value indicates a more stable factor space.

2.3.6 Economic Interpretability

The factor-loading shapes should be inspected over maturities M1–M24.

Typical patterns are:

- PC1: common level factor;
- PC2: front-versus-back slope factor;
- PC3: curvature or winter–summer factor;
- PC4: more localized seasonal or maturity-bucket deformation.

The interpretation should be verified empirically and should not be imposed ex ante.

2.4 Metrics for Evaluating the Fundamental Regression

Once the PCA representation is validated, the regression is evaluated on its ability to explain and reconstruct the curve out-of-sample.

2.4.1 Out-of-Sample Factor R^2

For each factor k ,

$$R_{k,\text{OOS}}^2 = 1 - \frac{\sum_t \left(f_{t,k}^{\text{obs}} - \hat{f}_{t,k} \right)^2}{\sum_t \left(f_{t,k}^{\text{obs}} - \bar{f}_{k,\text{train}} \right)^2}.$$

A factor may explain a large fraction of price variation while remaining difficult to predict from fundamentals. Therefore, explained variance and factor predictability must be evaluated separately.

2.4.2 Fundamental Curve RMSE

The full model-implied curve is

$$\hat{x}_t^{\text{fund}} = s_t + \mu_t + D_{\varepsilon,t} V_{t,K} \hat{f}_t.$$

Its out-of-sample RMSE is

$$\text{RMSE}_{\text{fund}} = \sqrt{\frac{1}{TN} \sum_t \sum_i (x_{t,i} - \hat{x}_{t,i}^{\text{fund}})^2}.$$

The same metric should also be reported by maturity.

2.4.3 Fundamental Spread RMSE

For each spread $i - j$,

$$e_{t,i,j}^{\text{fund,spread}} = (x_{t,i} - x_{t,j}) - (\hat{x}_{t,i}^{\text{fund}} - \hat{x}_{t,j}^{\text{fund}}).$$

Then

$$\text{RMSE}_{i,j}^{\text{fund,spread}} = \sqrt{\frac{1}{T} \sum_t (e_{t,i,j}^{\text{fund,spread}})^2}.$$

2.4.4 Residual Fundamental Content

The valuation residual is

$$d_t = x_t - \hat{x}_t^{\text{fund}}.$$

The residuals should not retain a strong systematic relationship with the same fundamental variables already used in the regression.

This may be tested through regressions of the form

$$d_{t,i} = \alpha_i + \beta_i^\top q_t + u_{t,i}.$$

Large and persistent residual exposures suggest that relevant fundamental information remains unexplained.

2.4.5 Residual Convergence

If the residual is interpreted as a potential mispricing, its economic value must be assessed through future relative returns.

For horizon h , define the convergence statistic

$$\text{IC}_h = \text{Corr}_{\text{cross-sec}} \left(-d_{t,i}, r_{t \rightarrow t+h,i}^{\text{relative}} \right).$$

A positive value indicates that contracts trading above the model-implied curve tend subsequently to underperform, while contracts below the implied curve tend to outperform.

The following should be reported:

- mean cross-sectional IC;
- HAC-adjusted t -statistic;
- IC by subperiod;
- IC by maturity bucket;
- portfolio performance after transaction costs.

2.5 Separating PCA Error from Regression Error

The total valuation error can be decomposed as

$$x_t - \hat{x}_t^{\text{fund}} = \underbrace{x_t - \hat{x}_t^{\text{PCA}}}_{\text{dimension-reduction error}} + \underbrace{\hat{x}_t^{\text{PCA}} - \hat{x}_t^{\text{fund}}}_{\text{factor-prediction error}}.$$

This decomposition is essential for diagnosis.

- A large PCA error indicates that K is too small or that the curve transformation is inadequate.
- A small PCA error but large regression error indicates that the selected factors are difficult to explain using the available fundamentals.
- Small reconstruction and regression errors but weak residual convergence indicate that the valuation residual is not necessarily tradable.

2.6 PCA Hyperparameters

The principal PCA hyperparameters are:

- rolling estimation-window length L ;
- fast exponential half-life h_f ;
- slow exponential half-life h_s ;
- number of retained factors K ;
- raw versus deseasonalized log-prices;
- seasonal estimation-window length H_{season} ;
- mean versus median seasonal estimator;
- idiosyncratic-volatility floor or shrinkage intensity.

2.6.1 Rolling Window Length

A reasonable grid is

$$L \in \{3, 5\} \text{ years of weekly observations.}$$

A shorter window adapts more rapidly to regime changes but produces noisier loadings. A longer window produces more stable factors but may mix structurally different TTF regimes.

2.6.2 Fast and Slow Half-Lives

The fast stage estimates recent idiosyncratic volatility, while the slow stage extracts stable factors.

A parsimonious grid is

$$(h_f, h_s) \in \{(8, 26), (13, 52), (26, 104)\} \text{ weeks,}$$

with

$$h_s > h_f.$$

These correspond approximately to:

- two months versus six months;
- one quarter versus one year;
- six months versus two years.

2.6.3 Number of Factors

The main specifications are

$$K \in \{3, 4\},$$

with $K = 5$ used as a diagnostic robustness check.

The three primary model variants are

Model A: raw log-prices, $K = 3$,

Model B: raw log-prices, $K = 4$,

and

Model C: deseasonalized log-prices, $K = 3$.

2.6.4 Seasonality Hyperparameters

For the deseasonalized specification, a reasonable initial choice is

$$H_{\text{season}} = 5 \text{ years},$$

with the historical median used as the baseline seasonal estimator.

Robustness checks may consider

$$H_{\text{season}} \in \{3, 5, 8\} \text{ years},$$

and

$$\text{aggregator} \in \{\text{mean}, \text{median}\}.$$

2.7 Hyperparameter Selection

Hyperparameters must be selected exclusively through walk-forward validation.

For each candidate specification:

$$\theta = (L, h_f, h_s, K, \text{seasonality specification}),$$

the complete weekly pipeline is rerun chronologically.

The parameters should not be chosen solely by explained variance.

The preferred specification should achieve a satisfactory balance between:

low PCA reconstruction error,
low calendar-spread reconstruction error,
stable factor loadings,
economically interpretable factors,
positive OOS factor R^2 ,
low fundamental-curve RMSE,
stable residual-convergence IC.

A practical validation score may be defined as a weighted combination:

$$\mathcal{L}(\theta) = \omega_1 \text{RMSE}_{\text{fund}} + \omega_2 \text{RMSE}_{\text{fund,spread}} + \omega_3 \bar{d}^{\text{subspace}} - \omega_4 \bar{R}_{\text{OOS}}^2 - \omega_5 \text{IC},$$

where the weights ω_j are fixed before evaluating the candidate models.

Alternatively, the metrics may be used sequentially:

1. discard PCA specifications with poor reconstruction or unstable loadings;
2. among the remaining specifications, select those with the best out-of-sample factor predictability;
3. among these models, prefer the specification with the best spread reconstruction and residual convergence.

This sequential approach is generally easier to interpret and less sensitive to arbitrary metric weights.

2.8 Recommended Initial Grid

A compact initial grid is

$$L \in \{3, 5\} \text{ years,}$$

$$(h_f, h_s) \in \{(8, 26), (13, 52), (26, 104)\} \text{ weeks,}$$

and

$$\text{curve specification} \in \{(\text{raw}, K = 3), (\text{raw}, K = 4), (\text{deseasonalized}, K = 3)\}.$$

This gives

$$2 \times 3 \times 3 = 18$$

primary PCA specifications.

The baseline deseasonalized model may initially use

$$H_{\text{season}} = 5 \text{ years}$$

and the median seasonal estimator. Alternative seasonal windows and the mean estimator should be treated as robustness checks rather than included in the main grid.

2.9 Summary

The recommended workflow is

Past Friday curves $\rightarrow$ rolling two-stage PCA $\rightarrow$ historical factor scores
 $\rightarrow$ regression on historical fundamentals $\rightarrow$ frozen weekly model
 $\rightarrow$ current Friday fundamentals $\rightarrow$ predicted factors $\rightarrow$ implied curve
 $\rightarrow$ comparison with settlement $\rightarrow$ valuation residual $\rightarrow$ weekly refit.

The PCA should first be judged through

explained variance + curve RMSE + spread RMSE + loading stability.

The fundamental regression should then be judged through

factor OOS R^2 + fundamental-curve RMSE + spread RMSE + residual convergence.

The final model should not merely reconstruct the observed curve. It should produce stable, interpretable factors and valuation residuals that contain economically meaningful information about relative future performance.